CASE STUDY 8: TEXT COMPRESSION USING CHARACTER RUNS

1. Problem Statement

Text Compression Using Character Runs is a simple method used to reduce repeated characters in a string. The program counts consecutive repeated characters and represents them using the character followed by its count.

For example, the input `aaabbcccc` is compressed as `a3b2c4`.

If a character appears only once, its count is also displayed. The program accepts a string and generates its compressed form.

2. Objectives

1. To understand string manipulation in Python.
2. To identify consecutive repeated characters.
3. To count repeated characters.
4. To generate a compressed string.
5. To understand the use of loops and conditional statements.

3. Assumptions

1. The input contains alphabetic characters only.
2. The input string is not empty.
3. Uppercase and lowercase letters are treated as different characters.
4. Only consecutive repeated characters are grouped together.
5. Each character is followed by its count, even when the count is 1.

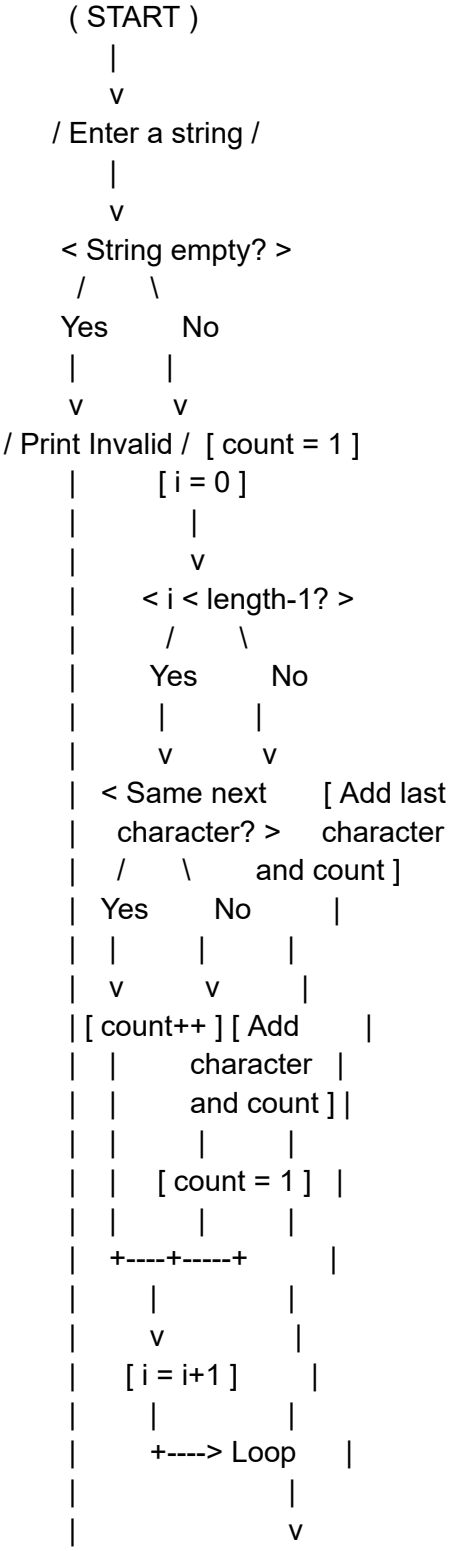
4. Algorithm

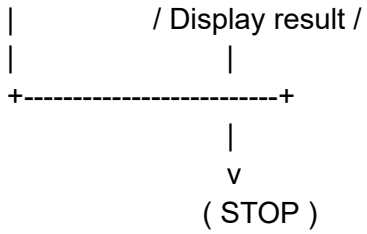
1. Start the program.
2. Read a string from the user.
3. Check whether the string is empty.
4. Initialize an empty compressed string.
5. Set the count of the first character to 1.
6. Compare each character with the next character.
7. If both characters are the same, increase the count.
8. Otherwise, add the current character and its count to the compressed string.
9. Reset the count to 1 for the new character.
10. Add the last character and its count to the compressed string.

11. Display the compressed string.
12. Stop the program.

5. Flowchart

Draw the following flowchart using standard symbols in your record.





6. Pseudocode

```
START
INPUT string

IF string is empty
   PRINT "Invalid input"
ELSE
   compressed = ""
   count = 1

   FOR i from 0 to length of string - 2
      IF string[i] == string[i+1]
         count = count + 1
      ELSE
         compressed = compressed + string[i] + count
         count = 1
      END IF
   END FOR

   compressed = compressed + last character + count
   PRINT compressed
END IF

STOP
```

7. Python Source Code

```
print("TEXT COMPRESSION USING CHARACTER RUNS")

text = input("Enter a string: ")

if text == "":
   print("Invalid input")
else:
   compressed = ""
   count = 1

   for i in range(len(text) - 1):
      if text[i] == text[i + 1]:
```

```
        count = count + 1
    else:
        compressed = compressed + text[i] + str(count)
        count = 1

compressed = compressed + text[-1] + str(count)

print("Compressed string:", compressed)
```

8. Test Cases and Expected Outputs

Test Case 1: Repeated characters

Input:

Enter a string: aaabbcccc

Expected Output:

TEXT COMPRESSION USING CHARACTER RUNS
Enter a string: aaabbcccc
Compressed string: a3b2c4

Explanation: The character a appears 3 consecutive times, b appears 2 times, and c appears 4 times.

Test Case 2: All characters are different

Input:

Enter a string: abcd

Expected Output:

TEXT COMPRESSION USING CHARACTER RUNS
Enter a string: abcd
Compressed string: a1b1c1d1

Explanation: Each character appears once, so the count of every character is 1.

Test Case 3: Empty input

Input:

Enter a string:

Expected Output:

TEXT COMPRESSION USING CHARACTER RUNS

Enter a string:
Invalid input

Explanation: The user has not entered any characters, so the program displays an invalid input message.

9. Output Screenshots

```
print("TEXT COMPRESSION USING CHARACTER RUNS")

text = input("Enter a string: ")

if text == "":
    print("Invalid input")
else:
    compressed = ""
    count = 1

    for i in range(len(text) - 1):
        if text[i] == text[i + 1]:
            count = count + 1
        else:
            compressed = compressed + text[i] + str(count)
            count = 1

    compressed = compressed + text[-1] + str(count)

    print("Compressed string:", compressed)
```

```
>>>
= RESTART: C:/Users/sirit/OneDrive/Desktop/Python-Activity-Based-Learning/case8.
py
TEXT COMPRESSION USING CHARACTER RUNS
Enter a string: aaabbcccc
Compressed string: a3b2c4
>>>
= RESTART: C:/Users/sirit/OneDrive/Desktop/Python-Activity-Based-Learning/case8.
py
TEXT COMPRESSION USING CHARACTER RUNS
Enter a string: abcd
Compressed string: a1b1c1d1
>>>
= RESTART: C:/Users/sirit/OneDrive/Desktop/Python-Activity-Based-Learning/case8.
py
TEXT COMPRESSION USING CHARACTER RUNS
Enter a string:
Invalid input
>>>
```

10. Conclusion

The Text Compression Using Character Runs program compresses a string by counting consecutive repeated characters. It uses strings, loops, conditional statements, and counters in Python. This program helps beginners understand string processing and a basic text compression technique.